

Chapter 40

1. The magnitude L of the orbital angular momentum $\vec{L}$ is given by Eq. 40-2: $L = \sqrt{\ell(\ell+1)}\hbar$. On the other hand, the components L_z are $L_z = m_\ell\hbar$, where $m_\ell = -\ell, \dots, \ell$. Thus, the semi-classical angle is $\cos\theta = L_z / L$. The angle is the smallest when $m = \ell$, or

$$\cos\theta = \frac{\ell\hbar}{\sqrt{\ell(\ell+1)}\hbar} \Rightarrow \theta = \cos^{-1}\left(\frac{\ell}{\sqrt{\ell(\ell+1)}}\right).$$

With $\ell = 5$, we have $\theta = \cos^{-1}(5/\sqrt{30}) = 24.1^\circ$.

2. For a given quantum number n there are n possible values of ℓ , ranging from 0 to $n-1$. For each ℓ the number of possible electron states is $N_\ell = 2(2\ell + 1)$. Thus the total number of possible electron states for a given n is

$$N_n = \sum_{\ell=0}^{n-1} N_\ell = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) = 2n^2.$$

Thus, in this problem, the total number of electron states is $N_n = 2n^2 = 2(5)^2 = 50$.

3. (a) We use Eq. 40-2:

$$L = \sqrt{\ell(\ell+1)}\hbar = \sqrt{3(3+1)}(1.055 \times 10^{-34} \text{ J}\cdot\text{s}) = 3.65 \times 10^{-34} \text{ J}\cdot\text{s}.$$

(b) We use Eq. 40-7: $L_z = m_\ell\hbar$. For the maximum value of L_z set $m_\ell = \ell$. Thus

$$[L_z]_{\max} = \ell\hbar = 3(1.055 \times 10^{-34} \text{ J}\cdot\text{s}) = 3.16 \times 10^{-34} \text{ J}\cdot\text{s}.$$

4. For a given quantum number n there are n possible values of ℓ , ranging from 0 to $n-1$. For each ℓ the number of possible electron states is $N_\ell = 2(2\ell + 1)$. Thus, the total number of possible electron states for a given n is

$$N_n = \sum_{\ell=0}^{n-1} N_\ell = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) = 2n^2.$$

(a) In this case $n = 4$, which implies $N_n = 2(4^2) = 32$.

(b) Now $n = 1$, so $N_n = 2(1^2) = 2$.

(c) Here $n = 3$, and we obtain $N_n = 2(3^2) = 18$.

(d) Finally, $n = 2 \rightarrow N_n = 2(2^2) = 8$.

5. (a) For a given value of the principal quantum number n , the orbital quantum number ℓ ranges from 0 to $n - 1$. For $n = 3$, there are three possible values: 0, 1, and 2.

(b) For a given value of ℓ , the magnetic quantum number m_ℓ ranges from $-\ell$ to $+\ell$. For $\ell = 1$, there are three possible values: -1 , 0 , and $+1$.

6. For a given quantum number ℓ there are $(2\ell + 1)$ different values of m_ℓ . For each given m_ℓ the electron can also have two different spin orientations. Thus, the total number of electron states for a given ℓ is given by $N_\ell = 2(2\ell + 1)$.

(a) Now $\ell = 3$, so $N_\ell = 2(2 \times 3 + 1) = 14$.

(b) In this case, $\ell = 1$, which means $N_\ell = 2(2 \times 1 + 1) = 6$.

(c) Here $\ell = 1$, so $N_\ell = 2(2 \times 1 + 1) = 6$.

(d) Now $\ell = 0$, so $N_\ell = 2(2 \times 0 + 1) = 2$.

7. (a) Using Table 40-1, we find $\ell = [m_\ell]_{\max} = 4$.

(b) The smallest possible value of n is $n = \ell_{\max} + 1 \geq \ell + 1 = 5$.

(c) As usual, $m_s = \pm \frac{1}{2}$, so two possible values.

8. (a) For $\ell = 3$, the greatest value of m_ℓ is $m_\ell = 3$.

(b) Two states ($m_s = \pm \frac{1}{2}$) are available for $m_\ell = 3$.

(c) Since there are 7 possible values for m_ℓ : $+3, +2, +1, 0, -1, -2, -3$, and two possible values for m_s , the total number of state available in the subshell $\ell = 3$ is 14.

9. **THINK** Knowing the value of ℓ , the orbital quantum number, allows us to determine the magnitudes of the angular momentum and the magnetic dipole moment.

EXPRESS The magnitude of the orbital angular momentum is

$$L = \sqrt{\ell(\ell+1)}\hbar.$$

Similarly, with $\vec{\mu}_{\text{orb}} = -\frac{e}{2m}\vec{L}$, the magnitude of $\vec{\mu}_{\text{orb}}$ is

$$\mu_{\text{orb}} = \frac{e\hbar}{2m}\sqrt{\ell(\ell+1)} = \mu_B,$$

where $\mu_B = e\hbar/2m$ is the Bohr magneton.

ANALYZE (a) For $\ell=3$, we have

$$L = \sqrt{\ell(\ell+1)}\hbar = \sqrt{3(3+1)}\hbar = \sqrt{12}\hbar.$$

So the multiple is $\sqrt{12} \approx 3.46$.

(b) The magnitude of the orbital dipole moment is

$$\mu_{\text{orb}} = \sqrt{\ell(\ell+1)}\mu_B = \sqrt{12}\mu_B.$$

So the multiple is $\sqrt{12} \approx 3.46$.

(c) The largest possible value of m_ℓ is $m_\ell = \ell = 3$.

(d) We use $L_z = m_\ell\hbar$ to calculate the z component of the orbital angular momentum. The multiple is $m_\ell = 3$.

(e) We use $\mu_z = -m_\ell\mu_B$ to calculate the z component of the orbital magnetic dipole moment. The multiple is $-m_\ell = -3$.

(f) We use $\cos\theta = m_\ell/\sqrt{\ell(\ell+1)}$ to calculate the angle between the orbital angular momentum vector and the z axis. For $\ell=3$ and $m_\ell=3$, we have $\cos\theta = 3/\sqrt{12} = \sqrt{3}/2$, or $\theta = 30.0^\circ$.

(g) For $\ell=3$ and $m_\ell=2$, we have $\cos\theta = 2/\sqrt{12} = 1/\sqrt{3}$, or $\theta = 54.7^\circ$.

(h) For $\ell=3$ and $m_\ell=-3$, $\cos\theta = -3/\sqrt{12} = -\sqrt{3}/2$, or $\theta = 150^\circ$.

LEARN Neither $\vec{L}$ nor $\vec{\mu}_{\text{orb}}$ can be measured in any way. We can, however, measure their z components.

10. (a) For $n = 3$ there are 3 possible values of ℓ : 0, 1, and 2.

(b) We interpret this as asking for the number of distinct values for m_ℓ (this ignores the multiplicity of any particular value). For each ℓ there are $2\ell + 1$ possible values of m_ℓ . Thus the number of possible m_ℓ 's for $\ell = 2$ is $(2\ell + 1) = 5$. Examining the $\ell = 1$ and $\ell = 0$ cases cannot lead to any new (distinct) values for m_ℓ , so the answer is 5.

(c) Regardless of the values of n , ℓ and m_ℓ , for an electron there are always two possible values of m_s : $\pm \frac{1}{2}$.

(d) The population in the $n = 3$ shell is equal to the number of electron states in the shell, or $2n^2 = 2(3^2) = 18$.

(e) Each subshell has its own value of ℓ . Since there are three different values of ℓ for $n = 3$, there are three subshells in the $n = 3$ shell.

11. **THINK** We can only measure one component of $\vec{L}$, say L_z , but not all three components.

EXPRESS Since $L^2 = L_x^2 + L_y^2 + L_z^2$, $\sqrt{L_x^2 + L_y^2} = \sqrt{L^2 - L_z^2}$. Replacing L^2 with $\ell(\ell+1)\hbar^2$ and L_z with $m_\ell\hbar$, we obtain

$$\sqrt{L_x^2 + L_y^2} = \hbar\sqrt{\ell(\ell+1) - m_\ell^2}.$$

ANALYZE For a given value of ℓ , the greatest that m_ℓ can be is ℓ , so the smallest that $\sqrt{L_x^2 + L_y^2}$ can be is $\hbar\sqrt{\ell(\ell+1) - \ell^2} = \hbar\sqrt{\ell}$. The smallest possible magnitude of m_ℓ is zero, so the largest $\sqrt{L_x^2 + L_y^2}$ can be is $\hbar\sqrt{\ell(\ell+1)}$. Thus,

$$\hbar\sqrt{\ell} \leq \sqrt{L_x^2 + L_y^2} \leq \hbar\sqrt{\ell(\ell+1)}.$$

LEARN Once we have chosen to measure $\vec{L}$ along the z axis, the x - and y -components cannot be measured with infinite certainty.

12. The angular momentum of the rotating sphere, $\vec{L}_{\text{sphere}}$, is equal in magnitude but in opposite direction to $\vec{L}_{\text{atom}}$, the angular momentum due to the aligned atoms. The number of atoms in the sphere is $N = \frac{N_A m}{M}$, where $N_A = 6.02 \times 10^{23} / \text{mol}$ is Avogadro's number and $M = 0.0558 \text{ kg/mol}$ is the molar mass of iron. The angular momentum due to the aligned atoms is

$$L_{\text{atom}} = 0.12N(m_s \hbar) = 0.12 \frac{N_A m \hbar}{M} \frac{1}{2}.$$

On the other hand, the angular momentum of the rotating sphere is (see Table 10-2 for I)

$$L_{\text{sphere}} = I\omega = \left(\frac{2}{5} mR^2 \right) \omega.$$

Equating the two expressions, the mass m cancels out and the angular velocity is

$$\begin{aligned} \omega &= 0.12 \frac{5N_A \hbar}{4MR^2} = 0.12 \frac{5(6.02 \times 10^{23} / \text{mol})(6.63 \times 10^{-34} \text{ J} \cdot \text{s} / 2\pi)}{4(0.0558 \text{ kg/mol})(2.00 \times 10^{-3} \text{ m})^2} \\ &= 4.27 \times 10^{-5} \text{ rad/s} \end{aligned}$$

13. **THINK** A gradient magnetic field gives rise to a magnetic force on the silver atom.

EXPRESS The force on the silver atom is given by

$$F_z = -\frac{dU}{dz} = -\frac{d}{dz}(-\mu_z B) = \mu_z \frac{dB}{dz}$$

where μ_z is the z component of the magnetic dipole moment of the silver atom, and B is the magnetic field. The acceleration is

$$a = \frac{F_z}{M} = \frac{\mu_z (dB/dz)}{M},$$

where M is the mass of a silver atom.

ANALYZE Using the data given in Sample Problem —“Beam separation in a Stern-Gerlach experiment,” we obtain

$$a = \frac{(9.27 \times 10^{-24} \text{ J/T})(1.4 \times 10^3 \text{ T/m})}{1.8 \times 10^{-25} \text{ kg}} = 7.2 \times 10^4 \text{ m/s}^2.$$

LEARN The deflection of the silver atom is due to the interaction between the magnetic dipole moment of the atom and the magnetic field. However, if the field is uniform, then $dB/dz = 0$, and the silver atom will pass the poles undeflected.

14. (a) From Eq. 40-19,

$$F = \mu_B \left| \frac{dB}{dz} \right| = (9.27 \times 10^{-24} \text{ J/T}) (1.6 \times 10^2 \text{ T/m}) = 1.5 \times 10^{-21} \text{ N}.$$

(b) The vertical displacement is

$$\Delta x = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{F}{m}\right)\left(\frac{l}{v}\right)^2 = \frac{1}{2}\left(\frac{1.5 \times 10^{-21} \text{ N}}{1.67 \times 10^{-27} \text{ kg}}\right)\left(\frac{0.80 \text{ m}}{1.2 \times 10^5 \text{ m/s}}\right)^2 = 2.0 \times 10^{-5} \text{ m}.$$

15. The magnitude of the spin angular momentum is

$$S = \sqrt{s(s+1)}\hbar = \sqrt{3/2}\hbar,$$

where $s = \frac{1}{2}$ is used. The z component is either $S_z = \hbar/2$ or $-\hbar/2$.

(a) If $S_z = +\hbar/2$ the angle θ between the spin angular momentum vector and the positive z axis is

$$\theta = \cos^{-1}\left(\frac{S_z}{S}\right) = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) = 54.7^\circ.$$

(b) If $S_z = -\hbar/2$, the angle is $\theta = 180^\circ - 54.7^\circ = 125.3^\circ \approx 125^\circ$.

16. (a) From Fig. 40-10 and Eq. 40-18,

$$\Delta E = 2\mu_B B = \frac{2(9.27 \times 10^{-24} \text{ J/T})(0.50 \text{ T})}{1.60 \times 10^{-19} \text{ J/eV}} = 58 \mu\text{eV}.$$

(b) From $\Delta E = hf$ we get

$$f = \frac{\Delta E}{h} = \frac{9.27 \times 10^{-24} \text{ J}}{6.63 \times 10^{-34} \text{ J}\cdot\text{s}} = 1.4 \times 10^{10} \text{ Hz} = 14 \text{ GHz}.$$

(c) The wavelength is

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8 \text{ m/s}}{1.4 \times 10^{10} \text{ Hz}} = 2.1 \text{ cm}.$$

(d) The wave is in the short radio wave region.

17. The total magnetic field, $B = B_{\text{local}} + B_{\text{ext}}$, satisfies $\Delta E = hf = 2\mu B$ (see Eq. 40-22). Thus,

$$B_{\text{local}} = \frac{hf}{2\mu} - B_{\text{ext}} = \frac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(34 \times 10^6 \text{ Hz})}{2(1.41 \times 10^{-26} \text{ J/T})} - 0.78 \text{ T} = 19 \text{ mT}.$$

18. We let $\Delta E = 2\mu_B B_{\text{eff}}$ (based on Fig. 40-10 and Eq. 40-18) and solve for B_{eff} :

$$B_{\text{eff}} = \frac{\Delta E}{2\mu_B} = \frac{hc}{2\lambda\mu_B} = \frac{1240 \text{ nm} \cdot \text{eV}}{2(21 \times 10^{-7} \text{ nm})(5.788 \times 10^{-5} \text{ eV/T})} = 51 \text{ mT}.$$

19. The energy of a magnetic dipole in an external magnetic field $\vec{B}$ is $U = -\vec{\mu} \cdot \vec{B} = -\mu_z B$, where $\vec{\mu}$ is the magnetic dipole moment and μ_z is its component along the field. The energy required to change the moment direction from parallel to antiparallel is $\Delta E = \Delta U = 2\mu_z B$. Since the z component of the spin magnetic moment of an electron is the Bohr magneton μ_B ,

$$\Delta E = 2\mu_B B = 2(9.274 \times 10^{-24} \text{ J/T})(0.200 \text{ T}) = 3.71 \times 10^{-24} \text{ J}.$$

The photon wavelength is

$$\lambda = \frac{c}{f} = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})}{3.71 \times 10^{-24} \text{ J}} = 5.35 \times 10^{-2} \text{ m}.$$

20. Using Eq. 39-20 we find that the lowest four levels of the rectangular corral (with this specific “aspect ratio”) are nondegenerate, with energies $E_{1,1} = 1.25$, $E_{1,2} = 2.00$, $E_{1,3} = 3.25$, and $E_{2,1} = 4.25$ (all of these understood to be in “units” of $h^2/8mL^2$). Therefore, obeying the Pauli principle, we have

$$E_{\text{ground}} = 2E_{1,1} + 2E_{1,2} + 2E_{1,3} + E_{2,1} = 2(1.25) + 2(2.00) + 2(3.25) + 4.25$$

which means (putting the “unit” factor back in) that the lowest possible energy of the system is $E_{\text{ground}} = 17.25(h^2/8mL^2)$. Thus, the multiple of $h^2/8mL^2$ is 17.25.

21. Because of the Pauli principle (and the requirement that we construct a state of lowest possible total energy), two electrons fill the $n = 1, 2, 3$ levels and one electron occupies the $n = 4$ level. Thus, using Eq. 39-4,

$$\begin{aligned} E_{\text{ground}} &= 2E_1 + 2E_2 + 2E_3 + E_4 \\ &= 2\left(\frac{h^2}{8mL^2}\right)(1)^2 + 2\left(\frac{h^2}{8mL^2}\right)(2)^2 + 2\left(\frac{h^2}{8mL^2}\right)(3)^2 + \left(\frac{h^2}{8mL^2}\right)(4)^2 \\ &= (2+8+18+16)\left(\frac{h^2}{8mL^2}\right) = 44\left(\frac{h^2}{8mL^2}\right). \end{aligned}$$

Thus, the multiple of $h^2/8mL^2$ is 44.

22. Due to spin degeneracy ($m_s = \pm 1/2$), each state can accommodate two electrons. Thus, in the energy-level diagram shown, two electrons can be placed in the ground state with energy $E_1 = 4(h^2/8mL^2)$, six can occupy the “triple state” with $E_2 = 6(h^2/8mL^2)$,

and so forth. With 11 electrons, the lowest energy configuration consists of two electrons with $E_1 = 4(h^2/8mL^2)$, six electrons with $E_2 = 6(h^2/8mL^2)$, and three electrons with $E_3 = 7(h^2/8mL^2)$. Thus, we find the ground-state energy of the 11-electron system to be

$$\begin{aligned} E_{\text{ground}} &= 2E_1 + 6E_2 + 3E_3 = 2\left(\frac{4h^2}{8mL^2}\right) + 6\left(\frac{6h^2}{8mL^2}\right) + 3\left(\frac{7h^2}{8mL^2}\right) \\ &= [(2)(4) + (6)(6) + (3)(7)]\left(\frac{h^2}{8mL^2}\right) = 65\left(\frac{h^2}{8mL^2}\right). \end{aligned}$$

The first excited state of the 11-electron system consists of two electrons with $E_1 = 4(h^2/8mL^2)$, five electrons with $E_2 = 6(h^2/8mL^2)$, and four electrons with $E_3 = 7(h^2/8mL^2)$. Thus, its energy is

$$\begin{aligned} E_{\text{1st excited}} &= 2E_1 + 5E_2 + 4E_3 = 2\left(\frac{4h^2}{8mL^2}\right) + 5\left(\frac{6h^2}{8mL^2}\right) + 4\left(\frac{7h^2}{8mL^2}\right) \\ &= [(2)(4) + (5)(6) + (4)(7)]\left(\frac{h^2}{8mL^2}\right) = 66\left(\frac{h^2}{8mL^2}\right). \end{aligned}$$

Thus, the multiple of $h^2/8mL^2$ is 66.

23. THINK With eight electrons, the ground-state energy of the system is the sum of the energies of the individual electrons in the system's ground-state configuration.

EXPRESS In terms of the quantum numbers n_x , n_y , and n_z , the single-particle energy levels are given by

$$E_{n_x, n_y, n_z} = \frac{h^2}{8mL^2} [n_x^2 + n_y^2 + n_z^2].$$

The lowest single-particle level corresponds to $n_x = 1$, $n_y = 1$, and $n_z = 1$ and is $E_{1,1,1} = 3(h^2/8mL^2)$. There are two electrons with this energy, one with spin up and one with spin down. The next lowest single-particle level is three-fold degenerate in the three integer quantum numbers. The energy is

$$E_{1,1,2} = E_{1,2,1} = E_{2,1,1} = 6(h^2/8mL^2).$$

Each of these states can be occupied by a spin up and a spin down electron, so six electrons in all can occupy the states. This completes the assignment of the eight electrons to single-particle states.

ANALYZE The ground state energy of the system is

$$E_{\text{gr}} = (2)(3)(h^2/8mL^2) + (6)(6)(h^2/8mL^2) = 42(h^2/8mL^2).$$

Thus, the multiple of $h^2/8mL^2$ is 42.

LEARN We summarize the ground-state configuration and the energies (in multiples of $h^2/8mL^2$) in the chart below:

n_x	n_y	n_z	m_s	energy
1	1	1	$-1/2, +1/2$	3 + 3
1	1	2	$-1/2, +1/2$	6 + 6
1	2	1	$-1/2, +1/2$	6 + 6
2	1	1	$-1/2, +1/2$	6 + 6
			total	42

24. (a) Using Eq. 39-20 we find that the lowest five levels of the rectangular corral (with this specific “aspect ratio”) have energies

$$E_{1,1} = 1.25, E_{1,2} = 2.00, E_{1,3} = 3.25, E_{2,1} = 4.25, E_{2,2} = 5.00$$

(all of these understood to be in “units” of $h^2/8mL^2$). It should be noted that the energy level we denote $E_{2,2}$ actually corresponds to two energy levels ($E_{2,2}$ and $E_{1,4}$; they are degenerate), but that will not affect our calculations in this problem. The configuration that provides the lowest system energy higher than that of the ground state has the first three levels filled, the fourth one empty, and the fifth one half-filled:

$$E_{\text{first excited}} = 2E_{1,1} + 2E_{1,2} + 2E_{1,3} + E_{2,2} = 2(1.25) + 2(2.00) + 2(3.25) + 5.00$$

which means (putting the “unit” factor back in) the energy of the first excited state is $E_{\text{first excited}} = 18.00(h^2/8mL^2)$. Thus, the multiple of $h^2/8mL^2$ is 18.00.

(b) The configuration that provides the next higher system energy has the first two levels filled, the third one half-filled, and the fourth one filled:

$$E_{\text{second excited}} = 2E_{1,1} + 2E_{1,2} + E_{1,3} + 2E_{2,1} = 2(1.25) + 2(2.00) + 3.25 + 2(4.25)$$

which means (putting the “unit” factor back in) the energy of the second excited state is

$$E_{\text{second excited}} = 18.25(h^2/8mL^2).$$

Thus, the multiple of $h^2/8mL^2$ is 18.25.

(c) Now, the configuration that provides the *next* higher system energy has the first two levels filled, with the next three levels half-filled:

$$E_{\text{third excited}} = 2E_{1,1} + 2E_{1,2} + E_{1,3} + E_{2,1} + E_{2,2} = 2(1.25) + 2(2.00) + 3.25 + 4.25 + 5.00$$

which means (putting the “unit” factor back in) the energy of the third excited state is $E_{\text{third excited}} = 19.00(h^2/8mL^2)$. Thus, the multiple of $h^2/8mL^2$ is 19.00.

(d) The energy states of this problem and Problem 40-22 are suggested below:

_____ third excited $19.00(h^2/8mL^2)$

_____ second excited $18.25(h^2/8mL^2)$

_____ first excited $18.00(h^2/8mL^2)$

_____ ground state $17.25(h^2/8mL^2)$

25. (a) Promoting one of the electrons (described in Problem 40-21) to a not-fully occupied higher level, we find that the configuration with the least total energy greater than that of the ground state has the $n = 1$ and 2 levels still filled, but now has only one electron in the $n = 3$ level; the remaining two electrons are in the $n = 4$ level. Thus,

$$\begin{aligned} E_{\text{first excited}} &= 2E_1 + 2E_2 + E_3 + 2E_4 \\ &= 2 \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \downarrow \right]_{10^2} + 2 \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \downarrow \right]_{20^2} + \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \right]_{30^2} + 2 \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \downarrow \right]_{40^2} \\ &= 2 + 8 + 9 + 32 \left[\frac{h^2}{8mL^2} \right] = 51 \left[\frac{h^2}{8mL^2} \right]. \end{aligned}$$

Thus, the multiple of $h^2/8mL^2$ is 51.

(b) Now, the configuration which provides the next higher total energy, above that found in part (a), has the bottom three levels filled (just as in the ground state configuration) and has the seventh electron occupying the $n = 5$ level:

$$\begin{aligned} E_{\text{second excited}} &= 2E_1 + 2E_2 + 2E_3 + E_5 \\ &= 2 \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \downarrow \right]_{10^2} + 2 \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \downarrow \right]_{20^2} + 2 \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \downarrow \right]_{30^2} + \left[\frac{h^2}{8mL^2} \right] \left[\uparrow \right]_{50^2} \\ &= 2 + 8 + 18 + 25 \left[\frac{h^2}{8mL^2} \right] = 53 \left[\frac{h^2}{8mL^2} \right]. \end{aligned}$$

Thus, the multiple of $h^2/8mL^2$ is 53.

(c) The third excited state has the $n = 1, 3, 4$ levels filled, and the $n = 2$ level half-filled:

$$\begin{aligned}
 E_{\text{third excited}} &= 2E_1 + E_2 + 2E_3 + 2E_4 \\
 &= 2 \left(\frac{h^2}{8mL^2} \right) (1)^2 + \left(\frac{h^2}{8mL^2} \right) (2)^2 + 2 \left(\frac{h^2}{8mL^2} \right) (3)^2 + 2 \left(\frac{h^2}{8mL^2} \right) (4)^2 \\
 &= (2 + 4 + 18 + 32) \left(\frac{h^2}{8mL^2} \right) = 56 \left(\frac{h^2}{8mL^2} \right).
 \end{aligned}$$

Thus, the multiple of $h^2/8mL^2$ is 56.

(d) The energy states of this problem and Problem 40-21 are suggested below:

_____	third excited	$56(h^2/8mL^2)$
=====	second excited	$53(h^2/8mL^2)$
_____	first excited	$51(h^2/8mL^2)$
=====	ground state	$44(h^2/8mL^2)$

26. The energy levels are given by

$$E_{n_x, n_y, n_z} = \frac{h^2}{8m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right) = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2).$$

The Pauli principle requires that no more than two electrons be in the lowest energy level (at $E_{1,1,1} = 3(h^2/8mL^2)$ with $n_x = n_y = n_z = 1$), but — due to their degeneracies — as many as six electrons can be in the next three levels,

$$\begin{aligned}
 E' &= E_{1,1,2} = E_{1,2,1} = E_{2,1,1} = 6(h^2/8mL^2) \\
 E'' &= E_{1,2,2} = E_{2,2,1} = E_{2,1,2} = 9(h^2/8mL^2) \\
 E''' &= E_{1,1,3} = E_{1,3,1} = E_{3,1,1} = 11(h^2/8mL^2).
 \end{aligned}$$

Using Eq. 39-21, the level above those can only hold two electrons:

$$E_{2,2,2} = (2^2 + 2^2 + 2^2)(h^2/8mL^2) = 12(h^2/8mL^2).$$

And the next higher level can hold as much as twelve electrons and has energy

$$E'''' = 14(h^2/8mL^2).$$

(a) The configuration that provides the lowest system energy higher than that of the ground state has the first level filled, the second one with one vacancy, and the third one with one occupant:

$$E_{\text{first excited}} = 2E_{1,1,1} + 5E' + E'' = 2(3) + 5(6) + 9$$

which means (putting the “unit” factor back in) the energy of the first excited state is

$$E_{\text{first excited}} = 45(h^2/8mL^2).$$

Thus, the multiple of $h^2/8mL^2$ is 45.

(b) The configuration that provides the next higher system energy has the first level filled, the second one with one vacancy, the third one empty, and the fourth one with one occupant:

$$E_{\text{second excited}} = 2E_{1,1,1} + 5E' + E'' = 2(3) + 5(6) + 11$$

which means (putting the “unit” factor back in) the energy of the second excited state is $E_{\text{second excited}} = 47(h^2/8mL^2)$. Thus, the multiple of $h^2/8mL^2$ is 47.

(c) Now, there are a couple of configurations that provide the *next* higher system energy. One has the first level filled, the second one with one vacancy, the third and fourth ones empty, and the fifth one with one occupant:

$$E_{\text{third excited}} = 2E_{1,1,1} + 5E' + E''' = 2(3) + 5(6) + 12$$

which means (putting the “unit” factor back in) the energy of the third excited state is $E_{\text{third excited}} = 48(h^2/8mL^2)$. Thus, the multiple of $h^2/8mL^2$ is 48. The other configuration with this same total energy has the first level filled, the second one with two vacancies, and the third one with one occupant.

(d) The energy states of this problem and Problem 40-25 are suggested below:

_____ third excited $48(h^2/8mL^2)$

_____ second excited $47(h^2/8mL^2)$

_____ first excited $45(h^2/8mL^2)$

_____ ground state $42(h^2/8mL^2)$

27. **THINK** The four quantum numbers (n, ℓ, m_ℓ, m_s) identify the quantum states of individual electrons in a multi-electron atom.

EXPRESS A lithium atom has three electrons. The first two electrons have quantum numbers $(1, 0, 0, \pm 1/2)$. All states with principal quantum number $n = 1$ are filled. The next lowest states have $n = 2$.

The orbital quantum number can have the values $\ell = 0$ or 1 and of these, the $\ell = 0$ states have the lowest energy. The magnetic quantum number must be $m_\ell = 0$ since this is the only possibility if $\ell = 0$. The spin quantum number can have either of the values $m_s = -\frac{1}{2}$ or $+\frac{1}{2}$. Since there is no external magnetic field, the energies of these two states are the same.

ANALYZE (a) Therefore, in the ground state, the quantum numbers of the third electron are either $n = 2, \ell = 0, m_\ell = 0, m_s = -\frac{1}{2}$ or $n = 2, \ell = 0, m_\ell = 0, m_s = +\frac{1}{2}$. That is, $(n, \ell, m_\ell, m_s) = (2, 0, 0, +1/2)$ and $(2, 0, 0, -1/2)$.

(b) The next lowest state in energy is an $n = 2, \ell = 1$ state. All $n = 3$ states are higher in energy. The magnetic quantum number can be $m_\ell = -1, 0$, or $+1$; the spin quantum number can be $m_s = -\frac{1}{2}$ or $+\frac{1}{2}$. Thus, $(n, \ell, m_\ell, m_s) = (2, 1, 1, +1/2), (2, 1, 1, -1/2), (2, 1, 0, +1/2), (2, 1, 0, -1/2), (2, 1, -1, +1/2)$ and $(2, 1, -1, -1/2)$.

LEARN No two electrons can have the same set of quantum numbers, as required by the Pauli exclusion principle.

28. For a given value of the principal quantum number n , there are n possible values of the orbital quantum number ℓ , ranging from 0 to $n - 1$. For any value of ℓ , there are $2\ell + 1$ possible values of the magnetic quantum number m_ℓ , ranging from $-\ell$ to $+\ell$. Finally, for each set of values of ℓ and m_ℓ , there are two states, one corresponding to the spin quantum number $m_s = -\frac{1}{2}$ and the other corresponding to $m_s = +\frac{1}{2}$. Hence, the total number of states with principal quantum number n is

$$N = 2 \sum_{\ell=0}^{n-1} (2\ell + 1).$$

Now

$$\sum_{\ell=0}^{n-1} 2\ell = 2 \sum_{\ell=0}^{n-1} \ell = 2 \frac{n}{2} (n-1) = n(n-1),$$

since there are n terms in the sum and the average term is $(n - 1)/2$. Furthermore,

$$\sum_{\ell=0}^{n-1} 1 = n.$$

Thus, $N = 2 \sum_{\ell=0}^{n-1} (2\ell+1) = 2n^2$.

29. The total number of possible electron states for a given quantum number n is

$$N_n = \sum_{\ell=0}^{n-1} N_{\ell} = 2 \sum_{\ell=0}^{n-1} (2\ell+1) = 2n^2.$$

Thus, if we ignore any electron-electron interaction, then with 110 electrons, we would have two electrons in the $n=1$ shell, eight in the $n=2$ shell, 18 in the $n=3$ shell, 32 in the $n=4$ shell, and the remaining 50 ($=110-2-8-18-32$) in the $n=5$ shell. The 50 electrons would be placed in the subshells in the order $s, p, d, f, g, h, \dots$ and the resulting configuration is $5s^2 5p^6 5d^{10} 5f^{14} 5g^{18}$. Therefore, the spectroscopic notation for the quantum number ℓ of the last electron would be g .

Note, however, when the electron-electron interaction is considered, the ground-state electronic configuration of darmstadtium actually is $[\text{Rn}]5f^{14}6d^97s^1$, where

$$[\text{Rn}]: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 4f^{14} 5s^2 5p^6 5d^{10} 6s^2 6p^6$$

represents the inner-shell electrons.

30. When a helium atom is in its ground state, both of its electrons are in the $1s$ state. Thus, for each of the electrons, $n=1$, $\ell=0$, and $m_{\ell}=0$. One of the electrons is spin up $m_s = +\frac{1}{2}$ while the other is spin down $m_s = -\frac{1}{2}$. Thus,

(a) the quantum numbers (n, ℓ, m_{ℓ}, m_s) for the spin-up electron are $(1, 0, 0, +1/2)$, and

(b) the quantum numbers (n, ℓ, m_{ℓ}, m_s) for the spin-down electron are $(1, 0, 0, -1/2)$.

31. The first three shells ($n=1$ through 3), which can accommodate a total of $2+8+18=28$ electrons, are completely filled. For selenium ($Z=34$) there are still $34-28=6$ electrons left. Two of them go to the $4s$ subshell, leaving the remaining four in the highest occupied subshell, the $4p$ subshell.

(a) The highest occupied subshell is $4p$.

(b) There are four electrons in the $4p$ subshell.

For bromine ($Z=35$) the highest occupied subshell is also the $4p$ subshell, which contains five electrons.

(c) The highest occupied subshell is $4p$.

(d) There are five electrons in the $4p$ subshell.

For krypton ($Z = 36$) the highest occupied subshell is also the $4p$ subshell, which now accommodates six electrons.

(e) The highest occupied subshell is $4p$.

(f) There are six electrons in the $4p$ subshell.

32. (a) The number of different m_ℓ 's is $2\ell + 1 = 3$, ($m_\ell = 1, 0, -1$) and the number of different m_s 's is 2, which we denote as $+1/2$ and $-1/2$. The allowed states are $(m_{\ell_1}, m_{s_1}, m_{\ell_2}, m_{s_2}) = (1, +1/2, 1, -1/2), (1, +1/2, 0, +1/2), (1, +1/2, 0, -1/2), (1, +1/2, -1, +1/2), (1, +1/2, -1, -1/2), (1, -1/2, 0, +1/2), (1, -1/2, 0, -1/2), (1, -1/2, -1, +1/2), (1, -1/2, -1, -1/2), (0, +1/2, 0, -1/2), (0, +1/2, -1, +1/2), (0, +1/2, -1, -1/2), (0, -1/2, -1, +1/2), (0, -1/2, -1, -1/2), (-1, +1/2, -1, -1/2)$. So, there are 15 states.

(b) There are six states disallowed by the exclusion principle, in which both electrons share the quantum numbers: $(m_{\ell_1}, m_{s_1}, m_{\ell_2}, m_{s_2}) = (1, +1/2, 1, +1/2), (1, -1/2, 1, -1/2), (0, +1/2, 0, +1/2), (0, -1/2, 0, -1/2), (-1, +1/2, -1, +1/2), (-1, -1/2, -1, -1/2)$. So, if the Pauli exclusion principle is not applied, then there would be $15 + 6 = 21$ allowed states.

33. The kinetic energy gained by the electron is eV , where V is the accelerating potential difference. A photon with the minimum wavelength (which, because of $E = hc/\lambda$, corresponds to maximum photon energy) is produced when all of the electron's kinetic energy goes to a single photon in an event of the kind depicted in Fig. 40-15. Thus, with $hc = 1240 \text{ eV} \cdot \text{nm}$,

$$eV = \frac{hc}{\lambda_{\min}} = \frac{1240 \text{ eV} \cdot \text{nm}}{0.10 \text{ nm}} = 1.24 \times 10^4 \text{ eV}.$$

Therefore, the accelerating potential difference is $V = 1.24 \times 10^4 \text{ V} = 12.4 \text{ kV}$.

34. With $hc = 1240 \text{ eV} \cdot \text{nm} = 1240 \text{ keV} \cdot \text{pm}$, for the K_α line from iron, the energy difference is

$$\Delta E = \frac{hc}{\lambda} = \frac{1240 \text{ keV} \cdot \text{pm}}{193 \text{ pm}} = 6.42 \text{ keV}.$$

We remark that for the hydrogen atom the corresponding energy difference is

$$\Delta E_{12} = -13.6 \text{ eV} \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = 10 \text{ eV}.$$

That this difference is much greater in iron is due to the fact that its atomic nucleus contains 26 protons, exerting a much greater force on the K - and L -shell electrons than that provided by the single proton in hydrogen.

35. **THINK** X-rays are produced when a solid target (silver in this case) is bombarded with electrons whose kinetic energies are in the keV range.

EXPRESS The wavelength is $\lambda_{\min} = hc / K_0$, where K_0 is the initial kinetic energy of the incident electron.

ANALYZE (a) With $hc = 1240 \text{ eV} \cdot \text{nm}$, we obtain

$$\lambda_{\min} = \frac{hc}{K_0} = \frac{1240 \text{ eV} \cdot \text{nm}}{35 \times 10^3 \text{ eV}} = 3.54 \times 10^{-2} \text{ nm} = 35.4 \text{ pm} .$$

(b) A K_α photon results when an electron in a target atom jumps from the L -shell to the K -shell. The energy of this photon is

$$E = 25.51 \text{ keV} - 3.56 \text{ keV} = 21.95 \text{ keV}$$

and its wavelength is

$$\lambda_{K\alpha} = hc / E = (1240 \text{ eV} \cdot \text{nm}) / (21.95 \times 10^3 \text{ eV}) = 5.65 \times 10^{-2} \text{ nm} = 56.5 \text{ pm} .$$

(c) A K_β photon results when an electron in a target atom jumps from the M -shell to the K -shell. The energy of this photon is $25.51 \text{ keV} - 0.53 \text{ keV} = 24.98 \text{ keV}$ and its wavelength is

$$\lambda_{K\beta} = (1240 \text{ eV} \cdot \text{nm}) / (24.98 \times 10^3 \text{ eV}) = 4.96 \times 10^{-2} \text{ nm} = 49.6 \text{ pm} .$$

LEARN Note that the cut-off wavelength $\lambda_{\min}$ is characteristic of the incident electrons, not of the target material.

36. (a) We use $eV = hc / \lambda_{\min}$ (see Eq. 40-23 and Eq. 38-4). With $hc = 1240 \text{ eV} \cdot \text{nm} = 1240 \text{ keV} \cdot \text{pm}$, the mean value of $\lambda_{\min}$ is

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1240 \text{ keV} \cdot \text{pm}}{50.0 \text{ keV}} = 24.8 \text{ pm} .$$

(b) The values of λ for the K_α and K_β lines do not depend on the external potential and are therefore unchanged.

37. Suppose an electron with total energy E and momentum p spontaneously changes into a photon. If energy is conserved, the energy of the photon is E and its momentum has magnitude E/c . Now the energy and momentum of the electron are related by

$$E^2 = (pc)^2 + (mc^2)^2 \Rightarrow pc = \sqrt{E^2 - (mc^2)^2} .$$

Since the electron has nonzero mass, E/c and p cannot have the same value. Hence, momentum cannot be conserved. A third particle must participate in the interaction, primarily to conserve momentum. It does, however, carry off some energy.

38. From the data given in the problem, we calculate frequencies (using Eq. 38-1), take their square roots, look up the atomic numbers (see Appendix F), and do a least-squares fit to find the slope: the result is 5.02×10^7 with the odd-sounding unit of a square root of a hertz. We remark that the least squares procedure also returns a value for the y -intercept of this statistically determined “best-fit” line; that result is negative and would appear on a graph like Fig. 40-17 to be at about -0.06 on the vertical axis. Also, we can estimate the slope of the Moseley line shown in Fig. 40-17:

$$\frac{(1.95 - 0.50)10^9 \text{ Hz}^{1/2}}{40 - 11} \approx 5.0 \times 10^7 \text{ Hz}^{1/2} .$$

39. **THINK** The frequency of an x-ray emission is proportional to $(Z - 1)^2$, where Z is the atomic number of the target atom.

EXPRESS The ratio of the wavelength λ_{Nb} for the K_α line of niobium to the wavelength λ_{Ga} for the K_α line of gallium is given by

$$\lambda_{\text{Nb}}/\lambda_{\text{Ga}} = (Z_{\text{Ga}} - 1)^2 / (Z_{\text{Nb}} - 1)^2 ,$$

where Z_{Nb} is the atomic number of niobium (41) and Z_{Ga} is the atomic number of gallium (31). Thus, $\lambda_{\text{Nb}}/\lambda_{\text{Ga}} = (30)^2 / (40)^2 = 9/16 \approx 0.563$.

LEARN The frequency of the K_α line is given by Eq. 40-26:

$$f = (2.46 \times 10^{15} \text{ Hz})(Z - 1)^2 .$$

40. (a) According to Eq. 40-26, $f \propto (Z - 1)^2$, so the ratio of energies is (using Eq. 38-2)

$$\frac{f}{f'} = \left(\frac{Z - 1}{Z' - 1} \right)^2 .$$

(b) We refer to Appendix F. Applying the formula from part (a) to $Z = 92$ and $Z' = 13$, we obtain

$$\frac{E}{E'} = \frac{f}{f'} = \left(\frac{Z - 1}{Z' - 1} \right)^2 = \left(\frac{92 - 1}{13 - 1} \right)^2 = 57.5 .$$

(c) Applying this to $Z = 92$ and $Z' = 3$, we obtain

$$\frac{E}{E'} = \left(\frac{92-1}{3-1} \right)^2 = 2.07 \times 10^3 .$$

41. We use Eq. 36-31, Eq. 39-6, and $hc = 1240 \text{ eV}\cdot\text{nm} = 1240 \text{ keV}\cdot\text{pm}$. Letting $2d \sin \theta = m\lambda = mhc / \Delta E$, where $\theta = 74.1^\circ$, we solve for d :

$$d = \frac{mhc}{2\Delta E \sin \theta} = \frac{(1)(1240 \text{ keV}\cdot\text{nm})}{2(8.979 \text{ keV} - 0.951 \text{ keV})(\sin 74.1^\circ)} = 80.3 \text{ pm} .$$

42. Using $hc = 1240 \text{ eV}\cdot\text{nm} = 1240 \text{ keV}\cdot\text{pm}$, the energy difference $E_L - E_M$ for the x-ray atomic energy levels of molybdenum is

$$\Delta E = E_L - E_M = \frac{hc}{\lambda_L} - \frac{hc}{\lambda_M} = \frac{1240 \text{ keV}\cdot\text{pm}}{63.0 \text{ pm}} - \frac{1240 \text{ keV}\cdot\text{pm}}{71.0 \text{ pm}} = 2.2 \text{ keV} .$$

43. (a) An electron must be removed from the K -shell, so that an electron from a higher energy shell can drop. This requires an energy of 69.5 keV. The accelerating potential must be at least 69.5 kV.

(b) After it is accelerated, the kinetic energy of the bombarding electron is 69.5 keV. The energy of a photon associated with the minimum wavelength is 69.5 keV, so its wavelength is

$$\lambda_{\min} = \frac{1240 \text{ eV}\cdot\text{nm}}{69.5 \times 10^3 \text{ eV}} = 1.78 \times 10^{-2} \text{ nm} = 17.8 \text{ pm} .$$

(c) The energy of a photon associated with the K_α line is $69.5 \text{ keV} - 11.3 \text{ keV} = 58.2 \text{ keV}$ and its wavelength is

$$\lambda_{K\alpha} = (1240 \text{ eV}\cdot\text{nm}) / (58.2 \times 10^3 \text{ eV}) = 2.13 \times 10^{-2} \text{ nm} = 21.3 \text{ pm} .$$

(d) The energy of a photon associated with the K_β line is

$$E = 69.5 \text{ keV} - 2.30 \text{ keV} = 67.2 \text{ keV}$$

and its wavelength is, using $hc = 1240 \text{ eV}\cdot\text{nm}$,

$$\lambda_{K\beta} = hc/E = (1240 \text{ eV}\cdot\text{nm}) / (67.2 \times 10^3 \text{ eV}) = 1.85 \times 10^{-2} \text{ nm} = 18.5 \text{ pm} .$$

44. (a) and (b) Let the wavelength of the two photons be λ_1 and $\lambda_2 = \lambda_1 + \Delta\lambda$. Then,

$$eV = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_1 + \Delta\lambda} \Rightarrow \lambda_1 = \frac{-(\Delta\lambda/\lambda_0 - 2) \pm \sqrt{(\Delta\lambda/\lambda_0)^2 + 4}}{2/\Delta\lambda} .$$

Here, $\Delta\lambda = 130 \text{ pm}$ and

$$\lambda_0 = hc/eV = 1240 \text{ keV}\cdot\text{pm}/20 \text{ keV} = 62 \text{ pm},$$

where we have used $hc = 1240 \text{ eV}\cdot\text{nm} = 1240 \text{ keV}\cdot\text{pm}$. We choose the plus sign in the expression for λ_1 (since $\lambda_1 > 0$) and obtain

$$\lambda_1 = \frac{-(130 \text{ pm}/62 \text{ pm} - 2) + \sqrt{(130 \text{ pm}/62 \text{ pm})^2 + 4}}{2/62 \text{ pm}} = 87 \text{ pm}.$$

The energy of the electron after its first deceleration is

$$K = K_i - \frac{hc}{\lambda_1} = 20 \text{ keV} - \frac{1240 \text{ keV}\cdot\text{pm}}{87 \text{ pm}} = 5.7 \text{ keV}.$$

(c) The energy of the first photon is $E_1 = \frac{hc}{\lambda_1} = \frac{1240 \text{ keV}\cdot\text{pm}}{87 \text{ pm}} = 14 \text{ keV}$.

(d) The wavelength associated with the second photon is

$$\lambda_2 = \lambda_1 + \Delta\lambda = 87 \text{ pm} + 130 \text{ pm} = 2.2 \times 10^2 \text{ pm}.$$

(e) The energy of the second photon is $E_2 = \frac{hc}{\lambda_2} = \frac{1240 \text{ keV}\cdot\text{pm}}{2.2 \times 10^2 \text{ pm}} = 5.7 \text{ keV}$.

45. The initial kinetic energy of the electron is $K_0 = 50.0 \text{ keV}$. After the first collision, the kinetic energy is $K_1 = 25 \text{ keV}$; after the second, it is $K_2 = 12.5 \text{ keV}$; and after the third, it is zero.

(a) The energy of the photon produced in the first collision is $50.0 \text{ keV} - 25.0 \text{ keV} = 25.0 \text{ keV}$. The wavelength associated with this photon is

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV}\cdot\text{nm}}{25.0 \times 10^3 \text{ eV}} = 4.96 \times 10^{-2} \text{ nm} = 49.6 \text{ pm}$$

where we have used $hc = 1240 \text{ eV}\cdot\text{nm}$.

(b) The energies of the photons produced in the second and third collisions are each 12.5 keV and their wavelengths are

$$\lambda = \frac{1240 \text{ eV}\cdot\text{nm}}{12.5 \times 10^3 \text{ eV}} = 9.92 \times 10^{-2} \text{ nm} = 99.2 \text{ pm}.$$

46. The transition is from $n = 2$ to $n = 1$, so Eq. 40-26 combined with Eq. 40-24 yields

$$f = \left[\frac{m_e e^4}{8 \epsilon_0^2 h^3} \right] \left[\frac{1}{1^2} - \frac{1}{2^2} \right] (Z-1)^2$$

so that the constant in Eq. 40-27 is

$$C = \sqrt{\frac{3m_e e^4}{32 \epsilon_0^2 h^3}} = 4.9673 \times 10^7 \text{ Hz}^{1/2}$$

using the values in the next-to-last column in the table in Appendix B (but note that the power of ten is given in the middle column).

We are asked to compare the results of Eq. 40-27 (squared, then multiplied by the accurate values of h/e found in Appendix B to convert to x-ray energies) with those in the table of K_α energies (in eV) given at the end of the problem. We look up the corresponding atomic numbers in Appendix F.

(a) For Li, with $Z = 3$, we have

$$E_{\text{theory}} = \frac{h}{e} C^2 (Z-1)^2 = \frac{6.6260688 \times 10^{-34} \text{ J} \cdot \text{s}}{1.6021765 \times 10^{-19} \text{ J/eV}} (4.9673 \times 10^7 \text{ Hz}^{1/2})^2 (3-1)^2 = 40.817 \text{ eV}.$$

The percentage deviation is

$$\text{percentage deviation} = 100 \left(\frac{E_{\text{theory}} - E_{\text{exp}}}{E_{\text{exp}}} \right) = 100 \left(\frac{40.817 - 54.3}{54.3} \right) = -24.8\% \approx -25\%.$$

In subsequent calculations, we use the steps outlined above.

(b) For Be, with $Z = 4$, the percentage deviation is -15% .

(c) For B, with $Z = 5$, the percentage deviation is -11% .

(d) For C, with $Z = 6$, the percentage deviation is -7.9% .

(e) For N, with $Z = 7$, the percentage deviation is -6.4% .

(f) For O, with $Z = 8$, the percentage deviation is -4.7% .

(g) For F, with $Z = 9$, the percentage deviation is -3.5% .

(h) For Ne, with $Z = 10$, the percentage deviation is -2.6% .

(i) For Na, with $Z = 11$, the percentage deviation is -2.0% .

(j) For Mg, with $Z = 12$, the percentage deviation is -1.5% .

Note that the trend is clear from the list given above: the agreement between theory and experiment becomes better as Z increases. One might argue that the most questionable step in Section 40-10 is the replacement $e^4 \rightarrow (Z-1)^2 e^4$ and ask why this could not equally well be $e^4 \rightarrow (Z-9)^2 e^4$ or $e^4 \rightarrow (Z-8)^2 e^4$. For large Z , these subtleties would not matter so much as they do for small Z , since $Z - \xi \approx Z$ for $Z \gg \xi$.

47. Let the power of the laser beam be P and the energy of each photon emitted be E . Then, the rate of photon emission is

$$R = \frac{P}{E} = \frac{P}{hc/\lambda} = \frac{P\lambda}{hc} = \frac{(5.0 \times 10^{-3} \text{ W})(0.80 \times 10^{-6} \text{ m})}{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(2.998 \times 10^8 \text{ m/s})} = 2.0 \times 10^{16} \text{ s}^{-1}.$$

48. The Moon is a distance $R = 3.82 \times 10^8 \text{ m}$ from Earth (see Appendix C). We note that the “cone” of light has apex angle equal to 2θ . If we make the small angle approximation (equivalent to using Eq. 36-14), then the diameter D of the spot on the Moon is

$$D = 2R\theta = 2R\left(\frac{1.22\lambda}{d}\right) = \frac{2(3.82 \times 10^8 \text{ m})(1.22)(600 \times 10^{-9} \text{ m})}{0.12 \text{ m}} = 4.7 \times 10^3 \text{ m} = 4.7 \text{ km}.$$

49. Let the range of frequency of the microwave be Δf . Then the number of channels that could be accommodated is

$$N = \frac{\Delta f}{10 \text{ MHz}} = \frac{(2.998 \times 10^8 \text{ m/s})\left[\frac{1}{450 \text{ nm}} - \frac{1}{650 \text{ nm}}\right]}{10 \text{ MHz}} = 2.1 \times 10^7.$$

The higher frequencies of visible light would allow many more channels to be carried compared with using the microwave.

50. From Eq. 40-29, $N_2/N_1 = e^{-(E_2 - E_1)/kT}$. We solve for T :

$$T = \frac{E_2 - E_1}{k \ln(N_1/N_2)} = \frac{3.2 \text{ eV}}{(1.38 \times 10^{-23} \text{ J/K}) \ln(2.5 \times 10^{15}/6.1 \times 10^{13})} = 1.0 \times 10^4 \text{ K}.$$

51. **THINK** The number of atoms in a state with energy E is proportional to $e^{-E/kT}$, where T is the temperature on the Kelvin scale and k is the Boltzmann constant.

EXPRESS Thus, the ratio of the number of atoms in the thirteenth excited state to the number in the eleventh excited state is

$$\frac{n_{13}}{n_{11}} = \frac{e^{-E_{13}/kT}}{e^{-E_{11}/kT}} = e^{-(E_{13}-E_{11})/kT} = e^{-\Delta E/kT},$$

where $\Delta E = E_{13} - E_{11}$ is the difference in the energies:

$$\Delta E = E_{13} - E_{11} = 2(1.2 \text{ eV}) = 2.4 \text{ eV}.$$

ANALYZE For the given temperature, $kT = (8.62 \times 10^{-2} \text{ eV/K})(2000 \text{ K}) = 0.1724 \text{ eV}$. Hence,

$$\frac{n_{13}}{n_{11}} = e^{-2.4/0.1724} = 9.0 \times 10^{-7}.$$

LEARN The 13th excited state has higher energy than the 11th excited state. Therefore, we expect fewer atoms to be in the 13th excited state.

52. The energy of the laser pulse is

$$E_p = P\Delta t = (2.80 \times 10^6 \text{ J/s})(0.500 \times 10^{-6} \text{ s}) = 1.400 \text{ J}.$$

Since the energy carried by each photon is

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(2.998 \times 10^8 \text{ m/s})}{424 \times 10^{-9} \text{ m}} = 4.69 \times 10^{-19} \text{ J},$$

the number of photons emitted in each pulse is

$$N = \frac{E_p}{E} = \frac{1.400 \text{ J}}{4.69 \times 10^{-19} \text{ J}} = 3.0 \times 10^{18} \text{ photons}.$$

With each atom undergoing stimulated emission only once, the number of atoms contributed to the pulse is also 3.0×10^{18} .

53. Let the power of the laser beam be P and the energy of each photon emitted be E . Then, the rate of photon emission is

$$R = \frac{P}{E} = \frac{P}{hc/\lambda} = \frac{P\lambda}{hc} = \frac{(2.3 \times 10^{-3} \text{ W})(632.8 \times 10^{-9} \text{ m})}{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(2.998 \times 10^8 \text{ m/s})} = 7.3 \times 10^{15} \text{ s}^{-1}.$$

54. According to Sample Problem — “Population inversion in a laser,” the population ratio at room temperature is $N_x/N_0 = 1.3 \times 10^{-38}$. Let the number of moles of the lasing material needed be n ; then $N_0 = nN_A$, where N_A is the Avogadro constant. Also $N_x = 10$. We solve for n :

$$n = \frac{N_x}{1.3 \times 10^{-38} N_A} = \frac{10}{1.3 \times 10^{-38} (6.02 \times 10^{23})} = 1.3 \times 10^{15} \text{ mol.}$$

55. (a) If t is the time interval over which the pulse is emitted, the length of the pulse is

$$L = ct = (3.00 \times 10^8 \text{ m/s})(1.20 \times 10^{-11} \text{ s}) = 3.60 \times 10^{-3} \text{ m.}$$

(b) If E_p is the energy of the pulse, E is the energy of a single photon in the pulse, and N is the number of photons in the pulse, then $E_p = NE$. The energy of the pulse is

$$E_p = (0.150 \text{ J}) / (1.602 \times 10^{-19} \text{ J/eV}) = 9.36 \times 10^{17} \text{ eV}$$

and the energy of a single photon is $E = (1240 \text{ eV} \cdot \text{nm}) / (694.4 \text{ nm}) = 1.786 \text{ eV}$. Hence,

$$N = \frac{E_p}{E} = \frac{9.36 \times 10^{17} \text{ eV}}{1.786 \text{ eV}} = 5.24 \times 10^{17} \text{ photons.}$$

56. Consider two levels, labeled 1 and 2, with $E_2 > E_1$. Since $T = -|T| < 0$,

$$\frac{N_2}{N_1} = e^{-(E_2 - E_1)/kT} = e^{-|E_2 - E_1|/(-k|T|)} = e^{|E_2 - E_1|/k|T|} > 1.$$

Thus, $N_2 > N_1$; this is population inversion. We solve for T :

$$T = -|T| = -\frac{E_2 - E_1}{k \ln(N_2/N_1)} = -\frac{2.26 \text{ eV}}{(8.62 \times 10^{-5} \text{ eV/K}) \ln(1 + 0.100)} = -2.75 \times 10^5 \text{ K.}$$

57. (a) We denote the upper level as level 1 and the lower one as level 2. From $N_1/N_2 = e^{-(E_2 - E_1)/kT}$ we get (using $hc = 1240 \text{ eV} \cdot \text{nm}$)

$$\begin{aligned} N_1 &= N_2 e^{-(E_1 - E_2)/kT} = N_2 e^{-hc/\lambda kT} = (4.0 \times 10^{20}) \exp \left[-\frac{1240 \text{ eV} \cdot \text{nm}}{(580 \text{ nm})(8.62 \times 10^{-5} \text{ eV/K})(300 \text{ K})} \right] \\ &= 5.0 \times 10^{-16} \ll 1, \end{aligned}$$

so practically no electron occupies the upper level.

(b) With $N_1 = 3.0 \times 10^{20}$ atoms emitting photons and $N_2 = 1.0 \times 10^{20}$ atoms absorbing photons, then the net energy output is

$$E = (N_1 - N_2) E_{\text{photon}} = (N_1 - N_2) \frac{hc}{\lambda} = (2.0 \times 10^{20}) \frac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s}) (2.998 \times 10^8 \text{ m/s})}{580 \times 10^{-9} \text{ m}}$$

$$= 68 \text{ J}.$$

58. For the n th harmonic of the standing wave of wavelength λ in the cavity of width L we have $n\lambda = 2L$, so $n\Delta\lambda + \lambda\Delta n = 0$. Let $\Delta n = \pm 1$ and use $\lambda = 2L/n$ to obtain

$$|\Delta\lambda| = \frac{\lambda|\Delta n|}{n} = \frac{\lambda}{n} = \lambda \left(\frac{\lambda}{2L} \right) = \frac{(533 \text{ nm})^2}{2(8.0 \times 10^7 \text{ nm})} = 1.8 \times 10^{-12} \text{ m} = 1.8 \text{ pm}.$$

59. For stimulated emission to take place, we need a long-lived state above a short-lived state in both atoms. In addition, for the light emitted by A to cause stimulated emission of B , an energy match for the transitions is required. The above conditions are fulfilled for the transition from the 6.9 eV state (lifetime 3 ms) to 3.9 eV state (lifetime 3 μs) in A , and the transition from 10.8 eV (lifetime 3 ms) to 7.8 eV (lifetime 3 μs) in B . Thus, the energy per photon of the stimulated emission of B is $10.8 \text{ eV} - 7.8 \text{ eV} = 3.0 \text{ eV}$.

60. (a) The radius of the central disk is

$$R = \frac{1.22 f \lambda}{d} = \frac{(1.22)(3.50 \text{ cm})(515 \text{ nm})}{3.00 \text{ mm}} = 7.33 \text{ } \mu\text{m}.$$

(b) The average power flux density in the incident beam is

$$\frac{P}{\pi d^2 / 4} = \frac{4(5.00 \text{ W})}{\pi(3.00 \text{ mm})^2} = 7.07 \times 10^5 \text{ W/m}^2.$$

(c) The average power flux density in the central disk is

$$\frac{(0.84)P}{\pi R^2} = \frac{(0.84)(5.00 \text{ W})}{\pi(7.33 \text{ } \mu\text{m})^2} = 2.49 \times 10^{10} \text{ W/m}^2.$$

61. (a) If both mirrors are perfectly reflecting, there is a node at each end of the crystal. With one end partially silvered, there is a node very close to that end. We assume nodes at both ends, so there are an integer number of half-wavelengths in the length of the crystal. The wavelength in the crystal is $\lambda_c = \lambda/n$, where λ is the wavelength in a vacuum and n is the index of refraction of ruby. Thus $N(\lambda/2n) = L$, where N is the number of standing wave nodes, so

$$N = \frac{2nL}{\lambda} = \frac{2(1.75)(0.0600 \text{ m})}{694 \times 10^{-9} \text{ m}} = 3.03 \times 10^5.$$

(b) Since $\lambda = c/f$, where f is the frequency, $N = 2nLf/c$ and $\Delta N = (2nL/c)\Delta f$. Hence,

$$\Delta f = \frac{c\Delta N}{2nL} = \frac{(2.998 \times 10^8 \text{ m/s})(0.0600 \text{ m})}{2(1.75)(0.0600 \text{ m})} = 1.43 \times 10^9 \text{ Hz}.$$

(c) The speed of light in the crystal is c/n and the round-trip distance is $2L$, so the round-trip travel time is $2nL/c$. This is the same as the reciprocal of the change in frequency.

(d) The frequency is

$$f = c/\lambda = (2.998 \times 10^8 \text{ m/s})/(694 \times 10^{-9} \text{ m}) = 4.32 \times 10^{14} \text{ Hz}$$

and the fractional change in the frequency is

$$\Delta f/f = (1.43 \times 10^9 \text{ Hz})/(4.32 \times 10^{14} \text{ Hz}) = 3.31 \times 10^{-6}.$$

62. The energy carried by each photon is

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(2.998 \times 10^8 \text{ m/s})}{694 \times 10^{-9} \text{ m}} = 2.87 \times 10^{-19} \text{ J}.$$

Now, the photons emitted by the Cr ions in the excited state can be absorbed by the ions in the ground state. Thus, the average power emitted during the pulse is

$$P = \frac{(N_1 - N_0)E}{\Delta t} = \frac{(0.600 - 0.400)(4.00 \times 10^{19})(2.87 \times 10^{-19} \text{ J})}{2.00 \times 10^{-6} \text{ s}} = 1.1 \times 10^6 \text{ J/s}$$

or $1.1 \times 10^6 \text{ W}$.

63. Due to spin degeneracy ($m_s = \pm 1/2$), each state can accommodate two electrons. Thus, in the energy-level diagram shown, two electrons can be placed in the ground state with energy $E_1 = 3(h^2/8mL^2)$, six can occupy the “triple state” with $E_2 = 6(h^2/8mL^2)$, and so forth. With 22 electrons in the system, the lowest energy configuration consists of two electrons with $E_1 = 3(h^2/8mL^2)$, six electrons with $E_2 = 6(h^2/8mL^2)$, six electrons with $E_3 = 9(h^2/8mL^2)$, six electrons with $E_4 = 11(h^2/8mL^2)$, and two electrons with $E_5 = 12(h^2/8mL^2)$. Thus, we find the ground-state energy of the 22-electron system to be

$$\begin{aligned}
E_{\text{ground}} &= 2E_1 + 6E_2 + 6E_3 + 6E_4 + 2E_5 \\
&= 2\left(\frac{3h^2}{8mL^2}\right) + 6\left(\frac{6h^2}{8mL^2}\right) + 6\left(\frac{9h^2}{8mL^2}\right) + 6\left(\frac{11h^2}{8mL^2}\right) + 2\left(\frac{12h^2}{8mL^2}\right) \\
&= [(2)(3) + (6)(6) + (6)(9) + (6)(11) + (2)(12)]\left(\frac{h^2}{8mL^2}\right) \\
&= 186\left(\frac{h^2}{8mL^2}\right).
\end{aligned}$$

Thus, the multiple of $h^2/8mL^2$ is 186.

64. (a) In the lasing action the molecules are excited from energy level E_0 to energy level E_2 . Thus the wavelength λ of the sunlight that causes this excitation satisfies

$$\Delta E = E_2 - E_0 = \frac{hc}{\lambda},$$

which gives (using $hc = 1240 \text{ eV}\cdot\text{nm}$)

$$\lambda = \frac{hc}{E_2 - E_0} = \frac{1240 \text{ eV}\cdot\text{nm}}{0.289 \text{ eV} - 0} = 4.29 \times 10^3 \text{ nm} = 4.29 \text{ }\mu\text{m}.$$

(b) Lasing occurs as electrons jump down from the higher energy level E_2 to the lower level E_1 . Thus the lasing wavelength λ' satisfies

$$\Delta E' = E_2 - E_1 = \frac{hc}{\lambda'},$$

which gives

$$\lambda' = \frac{hc}{E_2 - E_1} = \frac{1240 \text{ eV}\cdot\text{nm}}{0.289 \text{ eV} - 0.165 \text{ eV}} = 1.00 \times 10^4 \text{ nm} = 10.0 \text{ }\mu\text{m}.$$

(c) Both λ and λ' belong to the infrared region of the electromagnetic spectrum.

65. (a) Using $hc = 1240 \text{ eV}\cdot\text{nm}$,

$$\Delta E = hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = (1240 \text{ eV}\cdot\text{nm})\left(\frac{1}{588.995 \text{ nm}} - \frac{1}{589.592 \text{ nm}}\right) = 2.13 \text{ meV}.$$

(b) From $\Delta E = 2\mu_B B$ (see Fig. 40-10 and Eq. 40-18), we get

$$B = \frac{\Delta E}{2\mu_B} = \frac{2.13 \times 10^{-3} \text{ eV}}{2(5.788 \times 10^{-5} \text{ eV/T})} = 18 \text{ T}.$$

66. (a) The energy difference between the two states 1 and 2 was equal to the energy of the photon emitted. Since the photon frequency was $f = 1666 \text{ MHz}$, its energy was given by

$$hf = (4.14 \times 10^{-15} \text{ eV}\cdot\text{s})(1666 \text{ MHz}) = 6.90 \times 10^{-6} \text{ eV}.$$

Thus,

$$E_2 - E_1 = hf = 6.90 \times 10^{-6} \text{ eV} = 6.90 \text{ } \mu\text{eV}.$$

(b) The emission was in the *radio* region of the electromagnetic spectrum.

67. Letting $eV = hc/\lambda_{\min}$ (see Eq. 40-23 and Eq. 38-4), we get

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1240 \text{ nm}\cdot\text{eV}}{eV} = \frac{1240 \text{ pm}\cdot\text{keV}}{eV} = \frac{1240 \text{ pm}}{V}$$

where V is measured in kV.

68. (a) The distance from the Earth to the Moon is $d_{em} = 3.82 \times 10^8 \text{ m}$ (see Appendix C). Thus, the time required is given by

$$t = \frac{2d_{em}}{c} = \frac{2(3.82 \times 10^8 \text{ m})}{2.998 \times 10^8 \text{ m/s}} = 2.55 \text{ s}.$$

(b) We denote the uncertainty in time measurement as δt and let $2\delta d_{es} = 15 \text{ cm}$. Then, since $d_{em} \propto t$, $\delta t/t = \delta d_{em}/d_{em}$. We solve for δt :

$$\delta t = \frac{t\delta d_{em}}{d_{em}} = \frac{(2.55 \text{ s})(0.15 \text{ m})}{2(3.82 \times 10^8 \text{ m})} = 5.0 \times 10^{-10} \text{ s}.$$

(c) The angular divergence of the beam is

$$\theta = 2 \tan^{-1} \left(\frac{1.5 \times 10^3}{d_{em}} \right) = 2 \tan^{-1} \left(\frac{1.5 \times 10^3}{3.82 \times 10^8} \right) = (4.5 \times 10^{-4})^\circ.$$

69. **THINK** The intensity at the target is given by $I = P/A$, where P is the power output of the source and A is the area of the beam at the target. We want to compute I and compare the result with 10^8 W/m^2 .

EXPRESS The laser beam spreads because diffraction occurs at the aperture of the laser. Consider the part of the beam that is within the central diffraction maximum. The angular position of the edge is given by $\sin \theta = 1.22\lambda/d$, where λ is the wavelength and d is the diameter of the aperture. At the target, a distance D away, the radius of the beam is

$r = D \tan \theta$. Since θ is small, we may approximate both $\sin \theta$ and $\tan \theta$ by θ , in radians. Then,

$$r = D\theta = 1.22D\lambda/d.$$

ANALYZE (a) Thus, we find the intensity to be

$$I = \frac{P}{\pi r^2} = \frac{Pd^2}{\pi(1.22D\lambda)^2} = \frac{(5.0 \times 10^6 \text{ W})(4.0 \text{ m})^2}{\pi[1.22(3000 \times 10^3 \text{ m})(3.0 \times 10^{-6} \text{ m})]^2} = 2.1 \times 10^5 \text{ W/m}^2,$$

not great enough to destroy the missile.

(b) We solve for the wavelength in terms of the intensity and substitute $I = 1.0 \times 10^8 \text{ W/m}^2$:

$$\lambda = \frac{d}{1.22D} \sqrt{\frac{P}{\pi I}} = \frac{4.0 \text{ m}}{1.22(3000 \times 10^3 \text{ m})} \sqrt{\frac{5.0 \times 10^6 \text{ W}}{\pi(1.0 \times 10^8 \text{ W/m}^2)}} = 1.40 \times 10^{-7} \text{ m} = 140 \text{ nm}.$$

LEARN The wavelength corresponds to the x-rays on the electromagnetic spectrum.

70. (a) From Fig. 40-14 we estimate the wavelengths corresponding to the K_β line to be $\lambda_\beta = 63.0 \text{ pm}$. Using $hc = 1240 \text{ eV} \cdot \text{nm} = 1240 \text{ keV} \cdot \text{pm}$, we have

$$E_\beta = (1240 \text{ keV} \cdot \text{pm}) / (63.0 \text{ pm}) = 19.7 \text{ keV} \approx 20 \text{ keV}.$$

(b) For K_α , with $\lambda_\alpha = 70.0 \text{ pm}$, $E_\alpha = \frac{hc}{\lambda_\alpha} = \frac{1240 \text{ keV} \cdot \text{pm}}{70.0 \text{ pm}} = 17.7 \text{ keV} \approx 18 \text{ keV}$.

(c) Both Zr and Nb can be used, since $E_\alpha < 18.00 \text{ eV} < E_\beta$ and $E_\alpha < 18.99 \text{ eV} < E_\beta$. According to the hint given in the problem statement, Zr is the best choice.

(d) Nb is the second best choice.

71. The principal quantum number n must be greater than 3. The magnetic quantum number m_ℓ can have any of the values $-3, -2, -1, 0, +1, +2$, or $+3$. The spin quantum number can have either of the values $-\frac{1}{2}$ or $+\frac{1}{2}$.

72. For a given shell with quantum number n the total number of available electron states is $2n^2$. Thus, for the first four shells ($n = 1$ through 4) the numbers of available states are 2, 8, 18, and 32 (see Appendix G). Since $2 + 8 + 18 + 32 = 60 < 63$, according to the “logical” sequence the first four shells would be completely filled in an europium atom, leaving $63 - 60 = 3$ electrons to partially occupy the $n = 5$ shell. Two of these three electrons would fill up the $5s$ subshell, leaving only one remaining electron in the only partially filled subshell (the $5p$ subshell). In chemical reactions this electron would have the tendency to be transferred to another element, leaving the remaining 62 electrons in

chemically stable, completely filled subshells. This situation is very similar to the case of sodium, which also has only one electron in a partially filled shell (the $3s$ shell).

73. **THINK** One femtosecond (fs) is equal to 10^{-15} s.

EXPRESS The length of the pulse's wave train is given by $L = c\Delta t$, where Δt is the duration of the laser. Thus, the number of wavelengths contained in the pulse is

$$N = \frac{L}{\lambda} = \frac{c\Delta t}{\lambda}.$$

ANALYZE (a) With $\lambda = 500$ nm and $\Delta t = 10 \times 10^{-15}$ s, we have

$$N = \frac{L}{\lambda} = \frac{(3.0 \times 10^8 \text{ m/s})(10 \times 10^{-15} \text{ s})}{500 \times 10^{-9} \text{ m}} = 6.0.$$

(b) We solve for X from $10 \text{ fm}/1 \text{ m} = 1 \text{ s}/X$:

$$X = \frac{10 \times 10^{-15} \text{ m}}{1 \text{ s}} = \frac{10 \times 10^{-15} \text{ m}}{3.15 \times 10^7 \text{ s/y}} = 3.2 \times 10^6 \text{ y}.$$

LEARN Femtosecond lasers have important applications in areas such as micro-machining and optical data storage.

74. One way to think of the units of $\hbar$ is that, because of the equation $E = hf$ and the fact that f is in cycles/second, then the “explicit” units for $\hbar$ should be J·s/cycle. Then, since 2π rad/cycle is a conversion factor for cycles $\rightarrow$ radians, $\hbar = h/2\pi$ can be thought of as the Planck constant expressed in terms of radians instead of cycles. Using the precise values stated in Appendix B,

$$\begin{aligned} \hbar &= \frac{h}{2\pi} = \frac{6.62606876 \times 10^{-34} \text{ J}\cdot\text{s}}{2\pi} = 1.05457 \times 10^{-34} \text{ J}\cdot\text{s} = \frac{1.05457 \times 10^{-34} \text{ J}\cdot\text{s}}{1.6021765 \times 10^{-19} \text{ J/eV}} \\ &= 6.582 \times 10^{-16} \text{ eV}\cdot\text{s}. \end{aligned}$$

75. Without the spin degree of freedom the number of available electron states for each shell would be reduced by half. So the values of Z for the noble gas elements would become half of what they are now: $Z = 1, 5, 9, 18, 27$, and 43 . Of this set of numbers, the only one that coincides with one of the familiar noble gas atomic numbers ($Z = 2, 10, 18, 36, 54$, and 86) is 18 . Thus, argon would be the only one that would remain “noble.”

76. (a) The value of ℓ satisfies $\sqrt{\ell(\ell+1)}\hbar \approx \sqrt{\ell^2}\hbar = \ell\hbar = L$, so $\ell \simeq L/\hbar \simeq 3 \times 10^{74}$.

(b) The number is $2\ell + 1 \approx 2(3 \times 10^{74}) = 6 \times 10^{74}$.

(c) Since

$$\cos \theta_{\min} = \frac{m_{\ell \max} \hbar}{\sqrt{\ell(\ell+1)} \hbar} = \frac{1}{\sqrt{\ell(\ell+1)}} \approx 1 - \frac{1}{2\ell} = 1 - \frac{1}{2(3 \times 10^{74})}$$

or $\cos \theta_{\min} \approx 1 - \theta_{\min}^2/2 \approx 1 - 10^{-74}/6$, we have

$$\theta_{\min} \approx \sqrt{10^{-74}/3} = 6 \times 10^{-38} \text{ rad}.$$

The correspondence principle requires that all the quantum effects vanish as $\hbar \rightarrow 0$. In this case $\hbar/L$ is extremely small so the quantization effects are barely existent, with $\theta_{\min} \approx 10^{-38} \text{ rad} \approx 0$.

77. We use $eV = hc/\lambda_{\min}$ (see Eq. 40-23 and Eq. 38-4):

$$h = \frac{eV \lambda_{\min}}{c} = \frac{(1.60 \times 10^{-19} \text{ C})(40.0 \times 10^3 \text{ eV})(31.1 \times 10^{-12} \text{ m})}{2.998 \times 10^8 \text{ m/s}} = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}.$$

78. Using $hc = 1240 \text{ eV} \cdot \text{nm}$, we find the energy difference to be

$$\Delta E = hc \left(\frac{1}{\lambda_A} - \frac{1}{\lambda_B} \right) = (1240 \text{ eV} \cdot \text{nm}) \left(\frac{1}{500 \text{ nm}} - \frac{1}{510 \text{ nm}} \right) = 0.049 \text{ eV}.$$

79. (a) Using $E = -\partial V / \partial r$, we find the electric field to be

$$E = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r} \left[\frac{Ze}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3} \right) \right] = \frac{Ze}{4\pi\epsilon_0} \left(\frac{1}{r^2} - \frac{r}{R^3} \right)$$

(b) The electric field at $r = R$ vanishes: $E(r = R) = \frac{Ze}{4\pi\epsilon_0} \left(\frac{1}{R^2} - \frac{R}{R^3} \right) = 0$. Since $V = 0$ outside the sphere, we conclude that the electric field is zero in the region $r \geq R$.

(c) At $r = R$, the electric potential is

$$V(r = R) = \frac{Ze}{4\pi\epsilon_0} \left(\frac{1}{R} - \frac{3}{2R} + \frac{R^2}{2R^3} \right) = 0$$

The electric potential outside the sphere is also zero.